

Hein Zarni Aung

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EDUCATION

Stony Brook University — Expected Graduation: May 2027

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Honors & Scholarships: Global Excellence Scholarship

Relevant Coursework: Full Stack Development, Software Security, Systems Fundamentals I & II, Algorithms

RESEARCH

Undergraduate Research Assistant — Ethos Lab, SBU — Sep 2025 – Dec 2025 (paused)

PI: Abisheka Pitumpe | Team size: 2

- Built a real-time monitoring pipeline for newly issued SSL/TLS certificates using Selenium and Python to scrape, filter, and flag fraudulent domains, designing heuristic-based detection logic with a faculty PI.

INTERNSHIPS

Software Engineering Intern — Clustr — Jan 2026 – Sept 2026 (expected)

Cross-platform React Native social/scheduling app | ~8-person dev team

- Built and owned backend APIs using Firebase Cloud Functions — CRUD endpoints for communities, posts, and comments with enforced permission checks, join/leave workflows, and activity feed endpoints, shipped end-to-end into production for a cross-platform mobile app.
- Contributed 50+ commits across backend and mobile frontend features via pull requests, participating in code review and resolving merge conflicts with an 8-person team to keep community and scheduling modules in sync.
- Designed and implemented a mobile Scheduler feature (React Native) with tabbed navigation, a reschedule bottom sheet, custom date/time picker integration, and secure authentication (password reset flow, custom claims for onboarding).

Software Engineering Intern — Prenora Dynamics LLC — Jun 2026 – Sept 2026 (expected)

~8-person team

- Own end-to-end design decisions for an in-house, in-platform video chat feature involving client/server interactions, researching and evaluating available SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).

CAMPUS INVOLVEMENT

International Peer Mentor — Office of Global Affairs, Stony Brook University — Aug 2025 – Present

- Mentored 6 international students through biweekly one-on-one meetings, providing hands-on guidance on academic strategies, communication, and cross-cultural adjustment.

PROJECTS

TuneAcademy — Hack Brooklyn 2026 (36-hour hackathon) — Apr 2026

Winner: Best Prenora Dynamics Hack (Sponsor Challenge) | Team of 4

- Built a data pipeline extracting pitch, rhythm, and tone features from audio using Essentia/NumPy to power an AI/ML-driven matching feature pairing music students with instructors based on technical weaknesses.
- Pivoted from a planned machine learning model to rule-based scoring algorithms under hackathon time and compute constraints, delivering reliable weakness detection within the 36-hour build window.
- Collaborated in a 4-person team; broader app included a React/Tailwind frontend, Python/FastAPI backend in Docker, Cloud Firestore, GCP hosting, and ElevenLabs-generated voice feedback reports.

SKILLS

Languages: Python, TypeScript, JavaScript, Java, C

Backend: Node.js, Express.js, FastAPI

Databases & Cloud: PostgreSQL, MongoDB, Firebase/Firestore, AWS, GCP

Frontend: React, React Native, Next.js

Tools: Git, Docker, Linux, PM2, Selenium